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PI Name:	Rice, Kelly Ph.D.		
Project Title:	Quorum Sensing and S. aureus Virulence Potential During Simulated Microgravity Growth		
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Task Description:	Staphylococcus aureus can infect nearly every tissue and organ system in the human body, and community-acquired methicillin-resistant S. aureus (MRSA) strains have emerged in recent years; they are highly virulent and transmissible, infecting otherwise healthy individuals. The potential medical importance of this organism to astronaut health during space flight is illustrated by the fact that S. aureus was recovered from the nasal passages of Apollo 13 and 14 astronauts at significantly higher numbers compared to preflight samples collected from the same astronauts. Our differential expression analysis of NASA GeneLab RNA-Seq data (GLDS-145) from S. aureus grown in Biological Research In Canisters Petri Dish Fixation Units (BRIC-PDFUs) suggests that genes of the S. aureus accessory gene regulator (agr) quorum-sensing two-component system were the most highly induced in space flight BRIC cultures. Agr upregulates expression of secreted toxins and tissue-degrading enzymes, and represses cell surface adhesins, in a cell-density dependent manner. In space flight BRIC cultures, expression of the Agr effector RNAIII (an untranslated small RNA that controls transcription and translation of Agr-regulated genes) was 88-fold increased. Additionally, expression of known Agr-regulated genes was altered in the space flight samples, including genes encoding secreted proteases (7 to 8-fold increase), phenol-soluble modulins (6 to 8-fold increase), and select cell surface adhesin genes (repressed 2 to 6-fold). Interestingly, S. aureus growth in High Aspect Rotating Vessels (HARVs), a ground-based low-shear model of simulated microgravity, was also previously reported to result in much greater levels of clumping relative to normal-gravity control cultures. Taken together, these data support the hypothesis that S. aureus Agr quorum sensing is upregulated during space flight due to microgravity-induced changes, which may impact its virulence potential. In objective 1, we will perform PacBio sequencing and label-free proteomics on genomic DNA and culture supernatant BRIC-23 space flight and ground control samples, respectively. DNA sequencing data will be analyzed for mutations and methylation patterns, and assessed for their correlation with gene expression changes identified in the BRIC-23 RNA-Seq data. The secreted proteomics data will determine if the secreted proteome profile of the space flight samples demonstrates a stronger Agr response than the ground-based control samples. In objective 2, the influence of Agr in simulated microgravity and normal gravity HARV cultures will be assessed. Specifically, we will monitor growth properties, secreted proteone profiles, strength and timing of Agr induction, and RNA-Seq profiling. The results from these experiments will help answer "basic questions about how microorganisms respond to changes in gravity," and thus directly relate to Research Emphasis 1 (Microbiomes of the Built Environment of Spacecraft Element) of this solicitation. The focus of these proposed experiments is on the contribution of the Agr quorum-sensing/cell communication system to S. aureus growth and virulence potential under simulated microgravity conditions, which will enable us to better design follow-up space flight experiments. Therefore, this proposed research also relates to Research Emphasis 4 (Molecular and Cellular Biology Element) by assessing the "impact of space flight on cell to cell interactions and communication." Up to 30% of the human population are colonized with S. aureus. It is considered part of the normal human microbiota and extremely difficult to completely remove from the body. These experiments will contribute to a better understanding of the effects of space flight environmental conditions on S. aureus-human interactions, which will allow us to design effective countermeasures that will protect astronaut health during long-term space flight missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Effective protection of astronaut health during space flight missions is predicated on a thorough understanding of the effects of this environment on the virulence potential of bacterial pathogens. The Staphylococcus aureus Agr quorum sensing system is a global regulator of virulence and metabolic gene expression, as well as biofilm formation. Therefore the observation that this system is stimulated during space flight growth conditions suggests that it may impact the pathogenic threat of S. aureus to astronauts during long-term space flight missions. The research proposed in this application is directed at gaining a better understanding of the contribution of Agr quorum sensing to physiological adaptation and pathogenic potential of S. aureus in response to microgravity conditions experienced during space flight and in a ground-based simulated microgravity model. It is anticipated that the results of these proposed experiments will allow us to make more refined predictions as to which gene expression changes and virulence properties are likely to occur in S. aureus during space flight, and how this may impact its pathogenic potential and effect on astronaut health. It is also anticipated that, by using an established ground-based system (low-shear modelled microgravity growth) to simulate microgravity conditions, in which growth and environmental parameters can be carefully controlled, we will be able to uncover new knowledge regarding the relationship between the S. aureus quorum-sensing response to microgravity and its virulence potential. This will contribute new data and information to GeneLab, and will allow for more refined hypotheses and experimental design of future space flight experiments to test the effect of space flight on S. aureus quorum sensing and virulence potential.
	Research progress: Objective 1. Determine if Accessory Gene Regulator (Agr)-specific changes in S. aureus Biological Research in Canisters (BRIC)-23 spaceflight gene expression correlate with genomic DNA mutations/methylations and the secreted proteome. • BRIC-23 RNAseq data analysis: Co-I Carroll was responsible for performing the original data analysis of the BRIC-23 Staphylococcus aureus spaceflight RNAseq experiment (https://). In brief, fastQ data files from n=9 flight and n=9 ground-control samples were imported into CLC Genomics Workbench (Qiagen) for analysis. Ribosomal RNA reads were filtered out, and remaining reads mapped to the S. aureus MRSA252 genome (Genbank #BX571856.1). The UAMS-1 genome (strain used in BRIC-23 flight experiment) is not closed and therefore the updated Genbank genome file for MRSA252 that contains annotations for sRNAs (1) and annotations for critical virulence genes such as PSM 1-4 was used instead. A previously published analysis pipeline was followed (1-3) using the “RNA-Seq analysis” feature of CLC Genomics Workbench, with quantile normalization of data sets (4). Standard cutoffs (= 2-fold change, mean normalized expression value = 10 for both samples) were used to curate differential gene expression (DE) data, performed as described in (5). Student’s t-test was used to determine significance. • BRIC-23 cell proteomics data analysis: Co-I Edelmann identified and quantified the proteins from the generated raw data from the BRIC-23 (https://) using Proteome Discoverer. Briefly, tandem mass spectra were extracted, charge state deconvoluted, and deisotoped using Proteome Discoverer (Thermo Fisher Scientific). Tandem mass spectrometry (MS/MS) samples were analyzed by using the SEQUEST algorithm (Thermo Fisher) using available databases containing S. aureus proteins (Genbank #BX571856.1). All analyzed fractions were merged before the analysis. SEQUEST search parameters were as follows: two maximum

trypsin mis-cleavages, precursor mass tolerance of 10 ppm, fragment mass tolerance of 0.6 Da; static modifications were TMT six-plex/+229.163 Da (N-terminus, Lys) and carbamidomethyl modification/+57.021 Da (Cys); dynamic modification was oxidation modification/+15.995 Da (Met). Maximum dynamic modifications per peptide were four. High XCorr Confidence Thresholds were 1.2, 1.9, 2.3, and 2.6 for z=1, 2, 3, and >4, respectively. The maximum allowable delta Cn value was 0.05 and strict false discovery rate (FDR) was established at 0.01, where the validation was done using the q-value method and after prior decoy databank search. All the peptides with medium and high confidence were used to identify and quantify proteins. The reporter ions (i.e. m/z 126, 127N, 127C, 128N, 131) were identified where the most confident centroid was used and 10 ppm for reporter ion mass tolerance. The reporter ion values were normalized to control samples (128N). Proteins belonging to multiple protein groups were grouped into a single accession number and final ratios were reported. Fold changes were calculated between flight and ground samples, where the P-value was calculated using the Student's t-test ($P<0.05$), indicating proteins with significant changes in abundance (minimum 1.5-fold change).

• BRIC-23 secretomics and data analysis: A biospecimen request for the BRIC-23 spaceflight and ground control samples needed for this objective was submitted through the NASA Life Sciences Data Archive (LSDA) website, followed by submission of a required short proposal which had to be reviewed and approved. This took some time, and combined with the subsequent COVID shutdown of research activities on both ends, the shipping of these samples was delayed until the end of 2020. LSDA was not able to provide us with ground control cell pellets for the BRIC-23 experiment, so PacBio sequencing was not performed. This analysis will be instead done on a future Biological Research in Canisters-Petri Dish Fixation Units (BRIC-PDFU) S. aureus flight experiment that will be performed in conjunction with another NASA-funded project (Grant #: 80NSSC21K0601).

Once received, we performed secretomics analysis as follows: filter-sterilized and concentrated (5 kDa MW cutoff) supernatants were provided by LSDA (n=4 each of BRIC-23 spaceflight (FL) and ground control (GC) samples). These samples were precipitated with trichloroacetic acid (TCA), followed by acetone-washing of the precipitated protein pellets. These were then solubilized in a high molar urea buffer, and protein concentrations quantified using a Pierce BCA kit (Fisher). Proteins in each sample were then reduced by dithiothreitol and alkylated by iodoacetamide, followed by Sodium dodecyl-sulfate polyacrylamide gel electrophoresis (SDS-PAGE) and in-gel trypsin digestion to generate peptides (6). Protein content and quantification was performed using a label-free quantitative shotgun mass proteomics approach using a High Performance Liquid Chromatography (HPLC)-Orbitrap Fusion mass spectrometer (University of Florida Interdisciplinary Center for Biotechnology Research / UF-ICBR proteomics core). Co-I Edelmann identified and quantified the proteins from the generated raw data using Proteome Discoverer, as previously described in her publications (6-14). These data were also analyzed by Scaffold software (Proteome Science, USA) to identify secreted proteins with statistically significant alterations in expression. Briefly, tandem mass spectra were extracted, charge state deconvoluted, and deisotoped using Proteome Discoverer (Thermo Fisher Scientific). Tandem mass spectrometry (MS/MS) samples were analyzed by using the SEQUEST algorithm (Thermo Fisher) using available databases containing S. aureus proteins (Genbank #BX571856.1) and contaminants. Scaffold (Proteome Software, Inc., USA) was used to validate MS/MS-based peptide and protein identifications, where the required delta Cn scores will be >0.2 and XCorr scores were >1.2, 1.9, 2.3, and 2.6 for singly, doubly, triply, and quadruply charged peptides, respectively. Protein identifications were accepted if they were established at >95.0% probability and contained >2 identified peptides, with a peptide and protein FDR <0.1%. Protein probabilities were assigned by the Protein Prophet algorithm. Weighted spectral counts were used for protein quantification, and data normalized before fold changes were calculated. Fisher's exact test in conjunction with Benjamini-Hochberg multiple testing corrections were used to calculate statistical significance, with $P<0.05$ indicating proteins with significant changes in abundance (minimum 1.5-fold change).

• Results of BRIC-23 -omics analysis: Hierarchical clustering analysis showed that the biological replicates in each RNAseq experimental group (n=9 each) exhibited strong clustering, indicating consistency of the data across replicates in each group. For cellular proteomics, the metadata associated with this experiment in the GeneLab BRIC-23 data indicated that due to low protein yield from some of the ground-control samples, replicates had to be pooled prior to proteomics, resulting in only n=3 GC samples and n=9 FL samples. However, hierarchical clustering analysis on the FL and GC proteins that showed statistically significant differences in relative abundance (1.5-fold change) indicated good consistency and clustering between the biological replicates within each experimental group. Likewise, the n=4 flight and GC culture supernatant samples that we processed and subjected to secretomics analysis displayed strong clustering within each experimental group. TVenn analysis was performed on these data, revealing that 4 genes/proteins were overlapped between all 3 datasets, 20 genes/proteins overlapped between RNAseq and cellular proteomics, and 9 genes/proteins overlapped between RNAseq and secretomics. Analysis of the RNA-seq data revealed that RNAIII, the effector of the Agr quorum sensing system, was the most highly upregulated gene in spaceflight cultures (~88-fold) relative to GCs. Genes of the Agr operon (~14 fold) were also highly upregulated during spaceflight, followed by genes encoding secreted phenol-soluble modulins (PSMs) and secreted proteases, all of which are upregulated by Agr. Upregulated spaceflight genes/proteins also had functions related to urease activity, Ess secretion, and copper transport. In line with the BRIC-23 RNA-seq and cellular proteomics data, spaceflight supernatants contained significantly increased abundance of several known secreted virulence factors, including Agr-regulated proteases (SspA, SspB, ScpA, Aur), staphylococcal nuclease (Nuc), and EsxA, a small protein secreted by the type VII-like Ess secretion system. Collectively, these data suggest that S. aureus experiences increased quorum sensing and altered expression of virulence factors in response to the spaceflight environment that may impact its pathogenic potential.

Task Progress:

• PSM quantification efforts: The BRIC-23 flight data indicated that the S. aureus Agr system was upregulated during spaceflight. One of the strongest indicators of S. aureus Agr activity is production of the PSM toxin peptides. These five peptides are known virulence factors that contribute to S. aureus infection, and their production is intimately associated with the Agr system. Consequently, one objective of Co-I Carroll's research was to quantify PSM abundance in culture supernatants from the BRIC-23 samples. To do this they initially used a crude organic extraction procedure and then attempted to refine the analysis using a more quantitative mass spectrometry-based approach. o Organic extraction (butanol extraction). Previous work by the Carroll group has shown that due to their amphipathic nature, the PSM peptides can be isolated and extracted from S. aureus culture supernatants using butanol. The resulting extracts are free from contaminating proteins and can be further quantified/studied. It was the intention to use this procedure to isolate and examine the PSMs produced by S. aureus during spaceflight; however, it soon became evident that this procedure was unsuitable. When high quantities of PSMs are produced (as is likely the case in the BRIC-23 samples) the butanol used for extraction quickly becomes saturated with peptides and consequently not all PSM peptides are removed. This leads to incomplete extraction and an inaccurate estimation of PSM abundance. Since the BRIC-23 samples are of limited suppl\y, this procedure was deemed insufficient to accurately quantify PSM abundance, and therefore a more accurate quantification procedure was developed. o Mass spectrometry-based quantification. While the PSMs structural

properties make them ideally suited to extraction with organic solvents, these same properties also make them difficult to study using traditional proteomics approaches. To develop an accurate quantification procedure, Co-I Carroll has established a collaboration with the mass spectrometry laboratory core at the Cleveland Clinic. Synthetic PSM peptides were synthesized and used to develop a reverse phase HPLC coupled to a liquid chromatography mass spectrometry (LC-MS) procedure that allows each individual peptide to be identified and quantified. Once the procedure was established (for individual PSM peptides) the procedure was then applied to a mixed sample and all five peptides were identifiable from a mixed sample. This result suggests that the procedure can be used to identify/quantify PSMs from culture supernatants; however, thus far, attempts to do so from rich medium have been unsuccessful. The inability to detect PSMs from culture supernatants likely results from their lower concentration, compared to the synthetic PSMs used to develop the assay. Currently their lab is attempting to concentrate culture supernatants (without the use of butanol) in the hopes that concentrated supernatants will contain enough PSMs to detect/quantify via the reverse phase HPLC/LC-MS procedure.

Objective 2. Define the impact of Agr on the S. aureus transcriptome, secreted proteome, and phenotypic virulence properties in response to simulated microgravity growth. • We have generated data that supports an effect of Rotary Cell Culture System (RCCS) simulated microgravity growth on S. aureus physiology relative to RCCS normal gravity control growth, using the same growth media and temperature used in the BRIC-23 spaceflight experiment. A summary of these findings is as follows: (a) RCCS normal gravity control cultures dropped to a lower pH in stationary phase relative to RCCS simulated microgravity cultures; (b) Simulated microgravity cultures excreted increased levels of lactate and reduced levels of acetate relative to normal gravity controls; (c) Stationary phase simulated microgravity cells exhibited decreased carotenoid pigmentation relative to normal gravity cultures, a phenotype which has been previously-associated with S. aureus simulated microgravity growth (15); (d) Real-time polymerase chain reaction (PCR) analysis revealed that simulated microgravity cultures expressed elevated RNA levels of genes encoding anaerobic/fermentative enzymes (lactate dehydrogenases, nitrite reductase) relative to normal gravity cultures. Collectively, these results suggest that S. aureus undergoes fermentative metabolism when grown in low shear modeled microgravity (LSMMG) using the RCCS system, which correlates with a previously-published study of S. aureus LSMMG growth (15).

• Time-course analysis of RNAIII (the primary effector of Agr-based quorum sensing which controls transcription and/or translation of many Agr-regulated genes) promoter activity using a ß-galactosidase reporter plasmid in the RCCS revealed that simulated microgravity cultures had decreased Agr activity relative to normal gravity RCCS control cultures. Ongoing research efforts are geared at determining if oxygen availability is responsible for these differences, by growing control High Aspect Rotating Vessel (HARV) cultures in an inverted configuration, so that cells do not settle upon the oxygenation membrane of the HARV.

• Secretomics analysis of stationary phase LSMMG cultures revealed LSMMG samples had significantly decreased abundance of Agr-regulated virulence factors compared to normal gravity controls. This correlated with the RNAIII promoter reporter gene assays, which showed delayed and less strong induction of the Agr quorum sensing response in LSMMG. Collectively these results suggest that S. aureus metabolism may be similar in both spaceflight and LSMMG, characterized by increased lactate excretion and increased expression of fermentation genes. Although the expression profile of Agr and its target regulon was altered in opposite directions during spaceflight and LSMMG, these results demonstrate that this quorum sensing system is affected by as yet-unknown signals related to these environments.

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Bibliography Type: Description: (Last Updated: 02/19/2026)

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